# Zach Zukowski

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**AI GOVERNANCE · CONTROL-LAYER ACCOUNTABILITY · DILIGENCE FRAMEWORKS**

· Agent fleet: over 67,000 commits · typical overlap 14, peak 61 · commit gate of over 200 checks
· over 61,000 advisory firings measured across 37 classes · 78-skill library · 33-file canonical-state layer
· 42 USPTO patent applications across 32 families (Canonical-State Architecture for multi-agent AI research)
· 9 SSRN papers · 25 federal comment letters · FSB AI consultation response

## SUMMARY

Ten years in digital assets: independent protocol research from Jul 2016, institutional practice from Dec 2019. AI-augmented research and governance-infrastructure builder who turns control-layer and operator-accountability research into machine-readable diligence systems, now extending that accountability lens from stablecoin gateways to AI deployment. Founder of Tokenization Systems; first hire at Borderless Capital, an RIA.

Focus: research-infrastructure engineering, control-layer accountability, machine-readable diligence, agentic workflow automation · MCP server design, Monte Carlo, difference-in-differences, coalition-power analysis · Python, R, DuneSQL.

## EXPERIENCE

**Founder** · Tokenization Systems *Feb 2026 to Present*

- Built the human-in-the-loop layer over the fleet: a terminal grid of every live agent session with a phone control bar, and a web dashboard with live session control. Agents act inside declared authority boundaries; consent, outward-facing and irreversible actions stay with the human operator.
- Authored a nine-paper SSRN-live research program covering the Control Layer Intensity Index (CLII) for gateway concentration, the Minimum Viable Equivalence Pack (MVEP) for institutional tokenization diligence, programmable monetary-policy design (Adaptive Tokenomics), governance concentration, and operational risk in token economies.
- Contributed 25 comment letters across a federal regulatory program (OCC bank stablecoin engagement RIN 1557-AF41 plus supplemental control-layer extension; Treasury GENIUS Act state similarity; FDIC payment-stablecoin reserve standards RIN 3064-AG19 and stablecoin-subsidiary issuance approvals RIN 3064-AG20; FinCEN AML/CFT FINCEN-2026-0034; FinCEN-OFAC PPSI sanctions FINCEN-2026-0100; joint banking AML/CFT 91 FR 18304); first international standard-setter filing: FSB AI Sound Practices consultation response (Jul 2026), the first external deployment of the AI-domain extensions of the Minimum Viable Equivalence Pack and Control Layer Intensity Index (MVEP-AI / CLII-AI); Routing the Dollar submitted to the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); Governance Concentration published at Frontiers in Blockchain.
- Working dialogue with capital-markets infrastructure operators on stablecoin, deposit-token, tokenized-deposit, and tokenized-equity deployment.

**Senior Investment Analyst, Digital Assets** · Borderless Capital (RIA; first hire) *Dec 2019 to Feb 2026*

***Research Function & Automation***

- Built an automated SEC-filing and press-release monitoring pipeline (n8n + Apify) covering 30+ Digital Asset Treasury Stocks (DATs), tracking coins held, cash, debt, and share counts (float, adjusted, fully diluted) per issuer to compute multiple on NAV (mNAV) apples to apples across the sector; analyzed ~75 DATs in 2025 and invested in 18; AI-assisted diligence workflows reducing subscription-document review from ~4 hours to ~1 hour per deal.
- Built the firm's research function from scratch; led a four-analyst team and personally produced ~25 of its ~125 long-form publications annually; taught 100 teams across nine accelerator programs (2021 to 2025).

***Governance Standards Advisory***

- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.

***Investment & Conviction Calls***

- Sourced the **LayerZero seed round** investment, returning ~80x; the underlying cross-chain interoperability thesis preceded the firm's Cross-Chain Ecosystem Fund by two years.
- Predicted Thorchain's **$200M insolvency 13 months in advance**; Borderless held no exposure when the predicted failure mode materialized.

**Blockchain Research Analyst (Independent)** · Self-Employed *Jul 2016 to Present*

- Independent digital-asset research since Jul 2016, three years before joining Borderless Capital: reviewed 250+ protocol whitepapers, diligenced 1,000+ project and token-offering pitches, consulted 150+ projects on protocol design and tokenomics, and invested in 50+ token offerings.
- Hands-on validation across 30+ blockchains under live conditions: liquidity, fees, settlement finality, bridging, governance execution, and custody and security trade-offs. 30+ conferences and bank summits across the U.S., Europe, Asia, and Dubai.

**Analyst Intern** · Greenway Solutions (fraud-control consulting; client: USAA) *May to Sep 2016*

- Benchmarked fraud controls and the security-versus-friction tradeoff across online, mobile, and call-center channels at nine leading U.S. banks: authentication and password-reset controls, account-opening data requirements, and funds-availability and clearing timing; built the scoring framework and delivered the executive summary.

## SELECTED RESEARCH

- **Canonical-State Architecture.** Filed 41 US patent applications across 32 families in 2026. Twelve applications across four families cover coordination and integrity for concurrent AI-agent sessions on a shared version-controlled repository: per-session staging-index isolation, a multi-mode verification engine, a commit-time canonical-state coherence gate, recursive self-validation of discipline codifications, a tamper-evident coordination-proof log, authorship-narrowed commit replay with a per-writer edit-intent ledger, and statement-time provenance binding under fail-closed admission control; validated on over 67,000 commits and 12 to 55 concurrent agents over seven months (as of 2026-09-28), with a companion 73-page methodology preprint. The other twenty-eight families cover evidence-grounding interlocks against hallucinated machine assessment (two applications); typed-absence control attribution for pseudonymous on-chain systems (three); instrument-absence as a typed outcome with verifier-integrity controls (two); authority-aware context retrieval, resource-triggered session handoff, fleet launch-governance admission, contribution attestation, and derived-artifact coherence (six); calibration-gated detector promotion and evaluation instrumentation (three); reputation-weighted governance thresholds (two); and eleven families filed since August covering provenance-chain attribution, source-fidelity omission-and-fabrication detection, outbound-artifact gating as typed negative outcomes, blinded evaluation batteries, peer-health-gated destructive cleanup, confidence-gated typed-judgment offload, warranted destructive maintenance anchored to prior repository state, its receipt-gated key recovery, observer-first control arming, an operator-controlled agent perimeter, and governed agent authority (one each).
- **Auditing Governance Concentration Beyond Token Allocation.** Published at Frontiers in Blockchain (2026). Live-governance audit of 52 token protocols: launch allocation does not predict governance concentration; DePIN governance is more concentrated than DeFi after correction (Cohen's d = 0.65, a directionally robust medium effect).
- **Operational Risk in Token Economies.** Operational-risk taxonomy (74 risk vectors, 64 testable controls, 36 control objectives) mapped to a 624-configuration battery; Lido V3 staking-vault case study.
- **Adaptive Tokenomics.** Industrial control engineering (PID controller) applied to token-supply emission as programmable monetary-policy mechanism; 806-run agent-based simulation battery; cuts worst-case node-count variance ~6x under competitor-shock attack.
- **Who Burns the Tokens?** Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.
- **Routing the Dollar.** Submitted to Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways including Visa Direct-adjacent paths.
- **Minimum Viable Equivalence Pack.** Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at $40B scale. Editor-invited submission to the Journal of Financial Regulation.
- **The Control Layer of Agentic Payments.** First control-layer measurement of live agentic payments (x402): facilitator concentration above the antitrust highly-concentrated threshold on every chain measured, subsidy and proxy pipelines exposed by buyer cardinality, and a composition chain with no accountable layer; anchors the FSB AI-consultation response.
- **FSB AI Sound Practices response.** Filed 2026-07-01; the program's first international standard-setter filing. Three practice additions (composition-chain documentation, execution-authority evidence, dependency-concentration monitoring) plus the invited nonbank case study on settlement facilitators.
- **Seven Dollars.** Cross-product framework for major dollar-denominated digital products; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).

## TECHNICAL

**Methods:** policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design

**Data:** Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring

**Stack:** Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino, BigQuery), JavaScript / TypeScript; 22 MCP and API connections wrapping the data sources above plus GitHub; Claude.ai + Claude Code + OpenAI Codex + Cursor with 78-skill library and 33-file canonical-state architecture; Jev (TypeSafe) typed-judgment offload; local models via Ollama (IBM Granite, Qwen); n8n, Apify, Git / GitHub

## RECOGNITION

- AI-governance focus group at Foresight Institute Vision Weekend
- Uniswap Foundation governance committee (2025)
- ETHDenver pitch judge (2025)
- Introduction to Tokenomics article publicly endorsed by Binance founder CZ on day of publication

## EDUCATION

**High Point University** | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President *2018*
Lingnan University, Hong Kong (Exchange)